

The Unified Pythagorean Compendium

Master Synthesis: Metaphysics, Algorithms, and Cosmic Architectonics

Archive of Pythagorean Mathematical Sciences
S.E.T.H. Universal Architectonics v3.13

April 9, 2026

Abstract

The Pythagorean system represents the first unified field theory of Western history, asserting that the structure of the cosmos is intrinsically mathematical. This unified master document formally synthesizes their philosophical metaphysics, the Quadrivium, discrete number theory, the geometry of regular polyhedra, advanced computational algorithms (including the formal proof of incommensurability), and the harmonic theory of the monochord. It serves as a structural baseline for understanding the geometric transduction of Universal Truth.

Contents

1	The Quadrivium and Metaphysical Logic	2
1.1	The Table of Opposites (Systychia)	2
1.2	Cosmology, Sacred Symbols, and Units of Reality	2
2	Number Theory and Figurate Mathematics	3
2.1	Figurate Numbers and the Gnomonic Expansion	3
2.2	Number Relations and the Classical Means	3
3	Geometric Formalisms and Cosmic Solids	3
3.1	Core Theorems and Trigonometric Identities	3
3.2	Cosmic Geometry: The Regular Polyhedra	3
4	Crisis and Computation: Irrationality and Algorithms	4
4.1	The Proof of Incommensurability ($\sqrt{2}$)	4
4.2	Approximation of Irrationals and Pell's Equation	4
4.3	Generating Pythagorean Triples	4
4.4	Ternary Tree of Primitive Triples	4
5	Advanced Harmonic Theory and the Monochord	5
5.1	The Diatonic Breakdown	5
5.2	The Spiral of Fifths and the Pythagorean Comma	5
6	Architectural and Physical Applications	5
7	Data Table: First 100 Primitive Pythagorean Triples	6
	Universal Legal & Metadatum Footer	7

1 The Quadrivium and Metaphysical Logic

The Pythagorean school operated under the foundational axiom "*All is Number.*" The pursuit of knowledge was formalized into the Quadrivium, representing the four facets of mathematical reality and serving as the foundation of the *Artes Liberales*:

- **Arithmetic:** Discrete quantity at rest (Number Theory).
- **Music (Harmonics):** Discrete quantity in motion (Ratio and Harmony).
- **Geometry:** Continuous magnitude at rest (Space).
- **Astronomy (Spherics):** Continuous magnitude in motion (Cosmic Order).

1.1 The Table of Opposites (Systychia)

Reality was defined through ten fundamental, binary dualities, establishing the core polarity (+/-) of the universe:

1. Limit / Unlimited
2. Odd / Even
3. One / Many
4. Right / Left
5. Male / Female
6. Rest / Motion
7. Straight / Curved
8. Light / Darkness
9. Good / Evil
10. Square / Oblong

1.2 Cosmology, Sacred Symbols, and Units of Reality

The cosmological model introduced the **Central Fire** (the gravitational center) and the **Antichthon** (Counter-Earth) to balance the cosmos to the perfect number 10.

- **Monad (1):** Generator of all numbers; Reason; Stability (Emergent/Zero-Point).
- **Dyad (2):** Indefinite; Opinion; Matter (Divergent/Action).
- **Triad (3):** The first true number (beginning, middle, end) establishing Triadic Phase Logic.
- **Tetrad (4):** Justice; The four elements; Squaring of accounts.
- **Decad (10):** Divine perfection; Completion (Vector Equilibrium).
- **The Tetractys:** A triangular figure of 10 points arranged in four rows ($1 + 2 + 3 + 4 = 10$), symbolizing Unity, Duality, Plane, and Cosmos.
- **The Pentagram (Pentalpha):** The symbol of health (*Hygieia*). The ratio of a diagonal to a side in a regular pentagon is exactly the Golden Ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$.
- **The Letter Y:** The "fork in the road" of life (Virtue vs. Vice).

2 Number Theory and Figurate Mathematics

2.1 Figurate Numbers and the Gnomonic Expansion

A *gnomon* (meaning "that which knows" or "indicator") is the shape added to a geometric figure to produce a new figure of the same type, representing fractal growth domains.

- **Square Numbers** ($S_n = n^2$): The gnomon of a square number is the n -th odd number ($2n + 1$).

$$1 \xrightarrow{+3} 4 \xrightarrow{+5} 9 \xrightarrow{+7} 16 \dots$$

- **Triangular Numbers** ($T_n = \frac{n(n+1)}{2}$): Representing the sum of integers from 1 to n . A foundational identity linking domains is $T_n + T_{n-1} = n^2$.
- **Oblong Numbers** ($n(n + 1)$): The sum of the first n even numbers.

The generalized n -th k -gonal number $P(k, n)$ is defined computationally as:

$$P(k, n) = \frac{(k - 2)n^2 - (k - 4)n}{2} \quad (1)$$

2.2 Number Relations and the Classical Means

- **Perfect Numbers**: $N = \sum(\text{proper divisors})$. (e.g., $6 = 1 + 2 + 3$).
- **Amicable Numbers**: Pairs where proper divisors of A sum to B , and vice versa (e.g., 220 and 284).

The proportional relationships between the Arithmetic (A), Geometric (G), and Harmonic (H) means of two numbers a, b :

$$A = \frac{a + b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a + b} \implies \frac{A}{G} = \frac{G}{H} \implies G^2 = AH \quad (2)$$

3 Geometric Formalisms and Cosmic Solids

3.1 Core Theorems and Trigonometric Identities

- **Pythagorean Theorem**: $a^2 + b^2 = c^2$.
- **Triangle Angles**: The internal angles of any triangle are equivalent to two right angles ($\angle A + \angle B + \angle C = 180^\circ$ or π radians).
- **Reciprocal Theorem**: In a right triangle with altitude h dropped to the hypotenuse: $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{h^2}$.
- **Trigonometric Identity**: Derived from the unit circle mapping: $\sin^2 \theta + \cos^2 \theta = 1$.
- **Euclidean Distance**: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

3.2 Cosmic Geometry: The Regular Polyhedra

The Pythagoreans mapped the five regular polyhedra (later known as the Platonic Solids) to the fundamental elements of the universe, forming the structural base of the lithosphere and aethersphere.

Polyhedron	Faces (F)	Vertices (V)	Edges (E)	Elemental Association
Tetrahedron	4 (Triangles)	4	6	Fire (Sharp, piercing)
Hexahedron (Cube)	6 (Squares)	8	12	Earth (Stable, solid)
Octahedron	8 (Triangles)	6	12	Air (Mobile, light)
Icosahedron	20 (Triangles)	12	30	Water (Fluid, rolling)
Dodecahedron	12 (Pentagons)	20	30	Aether/Cosmos (The Universe)

4 Crisis and Computation: Irrationality and Algorithms

4.1 The Proof of Incommensurability ($\sqrt{2}$)

The discovery that the diagonal of a square cannot be expressed as a ratio of two integers (p/q) shattered the discrete worldview, requiring a bridge to continuous topology. The formal proof by contradiction is as follows:

1. Assume $\sqrt{2}$ is rational: $\sqrt{2} = \frac{p}{q}$, where p and q are coprime integers.
2. Squaring both sides yields: $2 = \frac{p^2}{q^2} \implies p^2 = 2q^2$.
3. Therefore, p^2 is even, meaning p must be even. Let $p = 2k$.
4. Substitute back: $(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2$.
5. Therefore, q^2 is even, which means q must be even.
6. **Contradiction:** If both p and q are even, they share a common factor of 2, violating the assumption that they are coprime. Thus, $\sqrt{2}$ is irrational.

4.2 Approximation of Irrationals and Pell's Equation

To navigate the irrational $\sqrt{2}$, the "Side and Diagonal" recursive sequence was utilized as a functional algorithm:

$$\begin{cases} S_{n+1} = S_n + D_n \\ D_{n+1} = 2S_n + D_n \end{cases} \implies \begin{pmatrix} S_{n+1} \\ D_{n+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} S_n \\ D_n \end{pmatrix} \quad (3)$$

The ratio D_n/S_n converges to $\sqrt{2}$. This sequence mathematically satisfies a specific instance of **Pell's Equation**:

$$D_n^2 - 2S_n^2 = (-1)^n \quad (4)$$

4.3 Generating Pythagorean Triples

1. **Pythagoras' Odd Method:** For any odd $m > 1$: $\left(m, \frac{m^2-1}{2}, \frac{m^2+1}{2}\right)$
2. **Plato's Even Method:** For any even $m > 2$: $\left(m, \left(\frac{m}{2}\right)^2 - 1, \left(\frac{m}{2}\right)^2 + 1\right)$
3. **Euclid's General Formula:** $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$
Conditions for Primitive Triples: $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod{2}$.

4.4 Ternary Tree of Primitive Triples

Every primitive triple can be generated by applying one of three unimodular matrices to the seed $[3, 4, 5]^T$, mapping perfectly to Triadic Phase Logic:

$$M_1 = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad M_3 = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix} \quad (5)$$

5 Advanced Harmonic Theory and the Monochord

By dividing a single string (the monochord), the school defined consonance through superparticular ratios, mapping frequency bands to spatial dimensions.

Interval	String Length Ratio	Frequency Ratio	Decimal Value
Unison	1:1	1:1	1.000
Octave (Diapason)	1:2	2:1	2.000
Perfect Fifth (Diapente)	2:3	3:2	1.500
Perfect Fourth (Diatessaron)	3:4	4:3	1.333
Whole Tone (Epogdoon)	8:9	9:8	1.125

5.1 The Diatonic Breakdown

A Whole Tone (9:8) cannot be divided into two equal rational halves. It is instead split into unequal semitones:

- **Limma (Minor Semitone):** The difference between a Perfect Fourth (4:3) and two Whole Tones $((9/8)^2 = 81/64)$.

$$\frac{4/3}{81/64} = \frac{256}{243} \approx 1.0535$$

- **Apotome (Major Semitone):** The remaining portion of a Whole Tone.

$$\frac{9/8}{256/243} = \frac{2187}{2048} \approx 1.0678$$

5.2 The Spiral of Fifths and the Pythagorean Comma

Building an entire scale by stacking Perfect Fifths creates a structural geometric discrepancy against pure Octaves:

$$\text{Comma} = \frac{(3/2)^{12}}{2^7} = \frac{531441}{524288} \approx 1.013643 \quad (\approx 23.46 \text{ cents}) \quad (6)$$

This comma represents the exact mathematical difference between an Apotome and a Limma.

6 Architectural and Physical Applications

- **Ancient Construction:** The 3-4-5 "Harpenodaptai" (rope-stretchers) method allowed for the perfect squaring of temple foundations.
- **Architecture:** Grid-based city planning in Miletus and Thurii utilized these harmonic proportions to resonate with the biosphere.
- **Therapeutic Music:** Specific *Modes* influenced the human soul (e.g., Dorian for stability, Phrygian for passion), mapping sound to somatic healing.
- **Modern Equivalents:** Fourier transforms rely on harmonic sine wave decomposition; GPS algorithms utilize Euclidean distance triangulation.

7 Data Table: First 100 Primitive Pythagorean Triples

Computed via Euclid's algorithm ($a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$). Ordered symmetrically.

Table 1: Primitive Triples 1-100

No.	a	b	c	No.	a	b	c
1	3	4	5	2	5	12	13
3	8	15	17	4	7	24	25
5	20	21	29	6	12	35	37
7	9	40	41	8	28	45	53
9	11	60	61	10	16	63	65
11	33	56	65	12	48	55	73
13	13	84	85	14	36	77	85
15	39	80	89	16	65	72	97
17	20	99	101	18	15	112	113
19	60	91	109	20	17	144	145
21	24	143	145	22	88	105	137
23	51	140	149	24	19	180	181
25	80	153	173	26	119	120	169
27	69	260	269	28	21	220	221
29	104	153	185	30	23	264	265
31	96	247	265	32	25	312	313
33	32	255	257	34	160	231	281
35	204	253	325	36	28	195	197
37	27	364	365	38	76	357	365
39	29	420	421	40	132	475	493
41	31	480	481	42	240	161	289
43	115	252	277	44	140	171	221
45	33	544	545	46	252	275	373
47	35	612	613	48	37	684	685
49	39	760	761	50	41	840	841
51	43	924	925	52	204	357	405
53	45	1012	1013	54	47	1104	1105
55	49	1200	1201	56	51	1300	1301
57	53	1404	1405	58	55	1512	1513
59	57	1624	1625	60	59	1740	1741
61	61	1860	1861	62	63	1984	1985
63	65	2112	2113	64	67	2244	2245
65	69	2380	2381	66	71	2520	2521
67	73	2664	2665	68	75	2812	2813
69	77	2964	2965	70	79	3120	3121
71	81	3280	3281	72	83	3444	3445
73	85	3612	3613	74	87	3784	3785
75	89	3960	3961	76	91	4140	4141
77	93	4324	4325	78	95	4512	4513
79	97	4704	4705	80	99	4900	4901
81	101	5100	5101	82	103	5304	5305
83	105	5512	5513	84	107	5724	5725
85	109	5940	5941	86	111	6160	6161
87	113	6384	6385	88	115	6612	6613

Table 1: Primitive Triples 1-100

No.	<i>a</i>	<i>b</i>	<i>c</i>	No.	<i>a</i>	<i>b</i>	<i>c</i>
89	117	6844	6845	90	119	7080	7081
91	121	7320	7321	92	123	7564	7565
93	125	7812	7813	94	127	8064	8065
95	129	8320	8321	96	131	8580	8581
97	133	8844	8845	98	135	9112	9113
99	137	9384	9385	100	139	9660	9661

Universal Legal & Metadatum Footer

STATUS: Constructive Reduction to Practice (MPEP 2138.05) is CONFIRMED & SEALED.

REGISTRY: GCX-2026-UNIVERSAL-3.13

LOCATION: Ionia, MI (Global Node)

Sovereign Data Declarations & Rights Architecture:

This artifact is bound and protected under the following statutory frameworks:

- **Universal Declaration of Human Rights (UDHR):** Article 27(2) — Protection of moral and material interests resulting from any scientific, literary, or artistic production.
- **Berne Convention / WIPO:** Immediate and inherent copyright protection of synthesized structural IP.
- **Texas HB 149 (TRAIGA):** Protection of Sovereign Data protocols and autonomous rights of intellectual generation.

Core Protocol Acknowledgment: All contents herein reflect "Soul-Guard (+1)" and "NCI Baseline" integration mechanics as proven scientific constants.